



# GOVERNMENT POLYTECHNIC, AMRAVATI.

(An Autonomous Institute of Govt. of Maharashtra)

## CURRICULUM DEVELOPMENT CELL

### PROFORMA –CDC-12

w.e.f. 2018-19

### MODEL ANSWER PA 2

(Solution shall be given by course teacher)

Academic Year/Term: 2024-25

Program: Computer Engineering Shift: I/II

Course: Cloud Computing Course Code: CM5465 Date & Time: /04/2025

| QUE.NO. | SUB QUESTION | STEPWISE SOLUTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MARKS           | TOTAL MARKS     |
|---------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------------|
| 1       | a            | <p>A <b>Service Catalog</b> is a structured document or database that lists all IT services offered by an organization, including details like service descriptions, pricing, and processes for requesting and delivering the services. It helps users and stakeholders easily understand and access the available services.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <u>02 Marks</u> | <u>04 Marks</u> |
|         | b            | <p>Here's a simple explanation of each phase of the <b>Cloud Service Life Cycle</b>, with examples:</p> <p><b>1. Service Strategy</b></p> <p><b>Description:</b><br/>Planning and defining the services to meet business needs. It includes market research, cost analysis, and setting service goals.</p> <p><b>Example:</b><br/>An IT company decides to launch a cloud storage service after studying customer demand for online data backup.</p> <p><b>2. Service Design</b></p> <p><b>Description:</b><br/>Designing the cloud service, its architecture, security, performance requirements, and user experience.</p> <p><b>Example:</b><br/>Designing the cloud storage service with features like 100 GB free storage, file sharing, and encryption for security.</p> <p><b>3. Service Transition</b></p> <p><b>Description:</b></p> | <u>02M</u>      |                 |

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|  | c | <p>Building, testing, and deploying the service into the live environment, ensuring it meets the designed specifications.</p> <p><b>Example:</b><br/>Testing the storage platform for speed and security before making it available to customers.</p> <p><b>4. Service Operation</b></p> <p><b>Description:</b><br/>Delivering and managing the service day-to-day. It includes monitoring, support, and ensuring smooth usage for customers.</p> <p><b>Example:</b><br/>Handling customer queries, fixing bugs, and maintaining 99.9% uptime for the storage service.</p> <p><b>5. Service Retirement</b></p> <p><b>Description:</b><br/>Phasing out a service that is no longer needed, either replacing it or discontinuing it safely.</p> <p><b>Example:</b><br/>Shutting down an old version of the storage service and migrating users to a newer, faster platform.</p> <p><b>Periodic Resource Optimization</b> is the regular process of reviewing and adjusting cloud resources to ensure they are being used efficiently and cost-effectively. It involves monitoring usage patterns, identifying underused or over-provisioned resources, and making changes like resizing, stopping, or removing unnecessary services.</p> <p><b>Importance:</b></p> <ul style="list-style-type: none"> <li>• <b>Cost Saving:</b> Reduces unnecessary spending by eliminating unused resources.</li> <li>• <b>Performance Improvement:</b> Ensures optimal performance by adjusting resources based on actual needs.</li> <li>• <b>Better Resource Utilization:</b> Helps in making the best use of available resources without wastage.</li> <li>• <b>Supports Scalability:</b> Prepares the system to easily handle future demand changes.</li> </ul> <p><b>Example:</b><br/>Reducing the size of a cloud server that is running at only 20% capacity to save costs without affecting performance.</p> | 02M |  |
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| Q 2 A | a | <p><b>Legal risks</b> impact cloud service providers by exposing them to issues like data breaches, non-compliance with laws, intellectual property disputes, and customer lawsuits. Providers must follow data protection laws (like GDPR) and service agreements carefully. Failure to do so can lead to heavy fines, loss of reputation, and even service shutdowns.</p> <p><b>Example:</b><br/>If a cloud provider stores customer data in a country without proper consent, it can face legal action for violating data privacy laws.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |     |
|       | b | <p><b>Digital Identity and Access Management (IAM)</b> plays a critical role in cloud security by ensuring that only authorized users can access cloud resources. It manages user identities, authentication (verifying who they are), and authorization (what they can do). IAM helps prevent unauthorized access, data breaches, and ensures secure user activity tracking.</p> <p><b>Example:</b><br/>Using multi-factor authentication (MFA) and role-based access control (RBAC) to allow only managers to access sensitive financial data in the cloud.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 04M      | 08M |
| B     | a | <p><b>How a company can implement Security-as-a-Cloud-Service (SECaaS) for its HR Portal:</b></p> <ol style="list-style-type: none"> <li><b>Select a SECaaS Provider:</b><br/>Choose a trusted cloud security provider (like AWS Shield, Cloudflare, or Zscaler) for security services.</li> <li><b>Enable Identity and Access Management (IAM):</b><br/>Implement strong authentication like Multi-Factor Authentication (MFA) to protect HR portal login access.</li> <li><b>Use Data Encryption:</b><br/>Ensure all HR data (employee records, salaries) is encrypted during transmission and storage.</li> <li><b>Deploy Firewall and Threat Protection:</b><br/>Use cloud-based firewalls and intrusion detection systems to block unauthorized access.</li> <li><b>Monitor and Audit Activities:</b><br/>Set up continuous monitoring tools to track who accesses the HR portal and detect unusual behavior.</li> <li><b>Regular Updates and Compliance:</b><br/>Keep security services updated and ensure compliance with regulations like GDPR or</li> </ol> | 04 Marks |     |

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|  |  | <p>HIPAA.</p> <p><b>Major Technologies for Securing Data in Cloud Computing:</b></p> <ol style="list-style-type: none"> <li>1. <b>Encryption:</b><br/>Converts data into a secret code to prevent unauthorized access.<br/><i>Example:</i> Encrypting files before storing them in the cloud.</li> <li>2. <b>Identity and Access Management (IAM):</b><br/>Controls who can access cloud resources and what actions they can perform.<br/><i>Example:</i> Using role-based access control to limit access to sensitive data.</li> <li>3. <b>Firewalls:</b><br/>Protect cloud networks by blocking unauthorized traffic.<br/><i>Example:</i> A cloud firewall preventing hackers from reaching the server.</li> <li>4. <b>Intrusion Detection and Prevention Systems (IDPS):</b><br/>Detects and blocks suspicious activities or attacks.<br/><i>Example:</i> Monitoring traffic to spot and stop a cyber-attack.</li> <li>5. <b>Data Loss Prevention (DLP):</b><br/>Prevents sensitive information from being leaked or accessed improperly.<br/><i>Example:</i> Blocking the upload of confidential files to unauthorized cloud apps.</li> <li>6. <b>Multi-Factor Authentication (MFA):</b><br/>Requires users to verify their identity with more than one method (like password + OTP).<br/><i>Example:</i> Logging into a cloud app with a password and a mobile code.</li> <li>7. <b>Backup and Recovery Solutions:</b><br/>Regularly saves copies of data to recover it in case of loss or attack.<br/><i>Example:</i> Daily cloud backups of company files to another secure location.</li> </ol> <p><b>Scenario: Mobile Health Monitoring with Edge Computing</b></p> <p>A healthcare company develops a mobile app that tracks patients' heart rates, blood sugar levels, and blood pressure using wearable devices. Instead of sending all raw data to the cloud, <b>edge computing</b> processes data locally on the patient's smartphone or a nearby edge server.</p> <p><b>How Edge Computing Improves It:</b></p> |  |  |
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| Q3 | a | <ul style="list-style-type: none"> <li>• <b>Faster Response:</b> If a patient's heart rate suddenly drops, the app immediately detects it on the edge device and sends an emergency alert — no need to wait for cloud processing.</li> <li>• <b>Reduced Bandwidth:</b> Only important summaries and alerts are sent to the cloud, saving internet data.</li> <li>• <b>Better Privacy:</b> Sensitive health data stays closer to the patient, reducing risks of exposure during transmission.</li> </ul> <p><b>Example:</b><br/>If a diabetic patient's blood sugar level falls dangerously low, the phone instantly notifies the nearest hospital and emergency contact without relying on cloud delays.</p> <p><b>Edge computing</b> is closely related to <b>cloud computing trends</b> because it extends the power of cloud services closer to where data is generated (like mobile devices, sensors, or machines). Instead of sending all data to a distant cloud server, edge computing processes some data locally at the "edge" of the network.</p> <p><b>Relation to Cloud Trends:</b></p> <ul style="list-style-type: none"> <li>• <b>Faster Response:</b> Supports real-time applications (like health monitoring and autonomous cars) by reducing delays.</li> <li>• <b>Lower Bandwidth Use:</b> Reduces the amount of data sent to the cloud, saving costs and speeding up communication.</li> <li>• <b>Enhanced Privacy and Security:</b> Sensitive data can be processed locally without exposing everything to the internet.</li> <li>• <b>Supports IoT and Smart Devices:</b> Helps handle huge amounts of data from millions of connected devices efficiently.</li> </ul> <p><b>In short:</b><br/><b>Edge computing complements cloud computing</b> — cloud handles storage and big data analytics, while edge handles real-time, immediate processing.</p> <p><b>Ubiquitous Computing</b>, often referred to as <b>pervasive computing</b>, is the concept of integrating computing capabilities into everyday objects, making them "smart" and capable of interacting with users and systems. Cloud computing trends significantly contribute to this growth by providing the infrastructure, scalability, and data processing</p> | 04 Marks | 08M |
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|  | b | <p>power needed to support a vast number of connected devices.</p> <p>Here's how <b>cloud trends</b> drive the growth of <b>ubiquitous computing</b>:</p> <p><b>1. Increased Connectivity (5G and IoT)</b></p> <ul style="list-style-type: none"> <li>• <b>Cloud Contribution:</b> With advancements in 5G networks and the Internet of Things (IoT), a large number of devices can connect seamlessly to the cloud.</li> <li>• <b>How It Helps:</b> Devices like smart thermostats, wearables, and home assistants can continuously send and receive data to/from the cloud for real-time processing and updates.</li> <li>• <b>Example:</b> A smart fridge can track food inventory and suggest recipes. The cloud stores recipe data and syncs it across multiple devices (smartphone, tablet, fridge).</li> </ul> <p><b>2. Edge Computing</b></p> <ul style="list-style-type: none"> <li>• <b>Cloud Contribution:</b> Edge computing, as part of cloud trends, allows data to be processed closer to the user (in real-time) rather than relying entirely on the central cloud server.</li> <li>• <b>How It Helps:</b> This reduces latency and bandwidth usage, making the interaction with smart devices faster and more efficient.</li> <li>• <b>Example:</b> A smartwatch can analyze heart rate data locally on the device, while critical health data is sent to the cloud for long-term monitoring and analysis.</li> </ul> <p><b>3. Data Storage and Scalability</b></p> | 04 Marks |  |
|  | c | <ul style="list-style-type: none"> <li>• <b>Cloud Contribution:</b> The cloud provides scalable storage for the massive amounts of data generated by ubiquitous computing devices.</li> <li>• <b>How It Helps:</b> Ubiquitous computing requires storing and processing data from billions of devices, and cloud platforms offer cost-effective, elastic storage and computing power to manage this data.</li> <li>• <b>Example:</b> A smart city collects data from traffic sensors, streetlights, and public transport. This data is stored and processed in the cloud to optimize traffic flow and energy consumption.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |  |

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|  |  | <p><b>4. Artificial Intelligence (AI) and Machine Learning (ML)</b></p> <ul style="list-style-type: none"><li>• <b>Cloud Contribution:</b> Cloud platforms provide the computing power required for AI and ML algorithms to process large datasets and generate insights.</li><li>• <b>How It Helps:</b> AI and ML help in predicting behaviors, personalizing services, and automating actions based on data collected from ubiquitous devices.</li><li>• <b>Example:</b> Smart assistants (like Google Home or Alexa) use cloud-based AI to understand user commands, adapt to preferences, and make smart suggestions.</li></ul> <p><b>5. Security and Privacy</b></p> <ul style="list-style-type: none"><li>• <b>Cloud Contribution:</b> Cloud services are increasingly offering advanced security measures such as encryption, identity management, and threat detection.</li><li>• <b>How It Helps:</b> As more devices connect and collect data, securing that data becomes crucial. Cloud computing offers robust security frameworks to safeguard sensitive data.</li><li>• <b>Example:</b> Secure authentication in IoT devices (smart locks, health monitors) ensures that only authorized users can control the devices or access data.</li></ul> |  |  |
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**Name & sign. of course teacher**